PaddleOCR-VL-Circuit: Built for Schematic Diagram Understanding

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Abstract

Automatic recognition of circuit schematic diagrams is a core challenge in Electronic Design Automation (EDA). This technical report presents CircuitOCR, a system based on PaddleOCR-VL-0.9B for circuit schematic OCR and netlist extraction. We construct the V5 Golden dataset of 1,857 schematics (500 synthetic + 1,357 real KiCad projects) with 100% visual-literal alignment. Through systematic LoRA ablation experiments, we identify the root cause of modality collapse—fine-tuning the vision-language projector destroys pre-trained alignment—and introduce the LLM-Only LoRA strategy that resolves it. The V10-Fixed model uses wide LoRA ($r = 16$, $\alpha = 32$, 5.7M params), fixes three critical training bugs (causal token double-shift, BPE boundary merging, `set_state_dict` silent failure in Paddle 3.1.0), and trains for 3 epochs (1,165 steps) in 43 minutes on an RTX 4060 (8 GB). Phase 1 evaluation on easy50-pure (44 samples) yields Component F1 = 0.2061 (4.5× over baseline 0.0455) and Token Recall = 0.1540 (96× over baseline 0.0016). Phase 2 topology evaluation reveals the honest capability: joint (refdes, value) pair F1 = 0.019—only $\sim$2% of components are fully correct, with 87% of values hallucinated. All code, weights, datasets, and a live demo are open-sourced. Detailed experimental results, annotated examples, and version history are provided in the Appendix.

Keywords Circuit Schematic · OCR · PaddleOCR-VL · LoRA Fine-tuning · Netlist Extraction · Modality Collapse

1 Introduction

Vision-Language Models (VLMs) can directly output structured text from images end-to-end, but their application to circuit schematic diagrams remains unexplored. No dedicated public benchmark exists for circuit schematic OCR. Existing datasets like Open Schematics [bshada, 2025] lack standardized evaluation protocols, while textbook-derived datasets suffer from SPICE-induced visual-to-logic annotation discrepancies.

The industrial need is substantial. The global semiconductor market exceeded $600 billion in 2025 (SEMI/WSTS). PCB reverse engineering outsourcing alone exceeds $500 million annually, with approximately 30% of time spent on schematic reconstruction. Use cases include hardware reverse engineering, legacy document digitization (1980s–2000s paper schematics), cross-tool migration, and the 500,000+ KiCad/EAGLE projects on GitHub lacking structured netlist exports.

Circuit schematic understanding involves two heterogeneous sub-tasks: (1) visual text recognition—extracting component designators (R1, C1), values (10k, 100nF), and net labels (VCC, GND); (2) topology graph tracing—inferring electrical connectivity from wires and net labels. Our evaluation protocol measures both dimensions: text NED for explicit recognition and component F1 for implicit structural understanding.

2 System Setup

We adopt PaddleOCR-VL-0.9B [Cui et al., 2025] (908M params), which integrates a NaViT-style dynamic vision encoder with the ERNIE-4.5-0.3B language model, supporting 109 languages. The unfine-tuned base model fails systematically on circuit schematics: it outputs generic document text (e.g., parameter tables), memorized hallucinations, or repetitive token sequences. These failures stem from pre-training data dominated by general documents with negligible exposure to engineering drawings.

3 V5 Golden Dataset

3.1 Legacy Issues and Design Principles

Earlier dataset versions (V1–V4) had three fatal flaws: (1) GT-image misalignment—labels from KiCad S-expression parsing included invisible pin numbers and hidden net names; (2) format mismatch—Open Schematics subsets used horizontal space-concatenated format (62.1% single-word lines vs. test set's 98.6%); (3) duplication noise—22,340 intra-line duplicate word pairs contaminating 24.3% of samples.

V5 Golden was rebuilt on two principles: (1) GT must align 100% with visible on-image text, using `draw.text()` synchronous recording for synthetic data and SVG stroked-text extraction for real KiCad data, with zero AI-generated labels; (2) format must match the test set, replacing mismatched subsets with train-real data (97.9% single-word lines), reducing duplication noise by 99.5% (22,340 $\rightarrow$ 107).

3.2 Data Sources and Statistics

Synthetic V3 (500 images). Random circuit topologies (5–30 components, 10 types) rendered as PNG, with labels from `draw.text()` logs [Zhang and Chen, 2026].

train-real (1,357 images). Real open-source KiCad projects from GitHub, rendered via kicad-cli SVG $\rightarrow$ cairosvg PNG, with GT from SVG stroked-text visible elements in one-word-per-line vertical format.

Table 1: Dataset Composition and Split

Split	Synthetic V3	train-real	Total	avg Lines
Training	450	1,104	1,554	32.1
Validation	50	171	171	33.4
Test (easy50)	–	–	44	–
Test (easy100)	–	–	89	–
Total	**500**	**1,357**	**1,857**	–

During training, we apply online augmentations: random JPEG compression (quality 85–100), random scaling (max_dim 168–384px), and random horizontal flipping.

4 Model Fine-Tuning

4.1 Modality Collapse: Discovery and Root Cause

Initial full LoRA experiments ($r = 8$, q/k/v/o self-attention, 2.75M params on 2,433 samples) reduced NED from 0.8848 to 0.8554, but output diversity collapsed to 4%—nearly all test samples produced identical output. We define this as **Modality Collapse**.

Through six controlled experiments (E1–E6) summarized in Table 2, we systematically identified the root cause:

Core finding: Projector LoRA is the root cause. The vision-language projector (`mlp_AR`, 25.9M params) is the sole bridge from visual features to language space. LoRA perturbation distorts these features into out-of-distribution vectors the LLM cannot interpret, causing it to degenerate into repeating high-frequency training tokens. Freezing the Projector while teaching only the LLM to reinterpret existing visual features is the safe strategy.

Table 2: Ablation Experiments for Modality Collapse

Exp	Variable	Result	Conclusion
E1	Real vs. blank image	Outputs differ	Vision encoder works
E2	Resolution 168→336px	Still collapses	Resolution not the cause
E3	1 vs. 3 epochs	NED improves 0.9%	Multi-epoch helps but insufficient
E4	Add Projector LoRA	NED drops to 0.8271	**Projector is the bottleneck**
E5	Rank $r = 8 \to r = 16$	NED drops to 0.7961	Capacity helps, diversity still 4%
E6	Freeze Projector	Diversity 90%	**Collapse fully resolved**

4.2 V5: LLM-Only LoRA Validation

V5 freezes the Projector and fine-tunes only LLM self-attention ($r = 8$, $\alpha = 16$, 1.25M params), with resolution increased to 384px and learning rate 2×10^{-5}. Training on 1,857 samples for 1 epoch (928 steps, 22 min) achieved 90% output diversity—the first fine-tuned model to produce different outputs for different images. However, $r = 8$ capacity was insufficient: NED (0.9031) remained below baseline due to premature convergence at $\sim$200 steps.

4.3 V10-Fixed: Three Technical Fixes

V10-Fixed scales LoRA to $r = 16$, $\alpha = 32$ (5.7M params, 0.63% of 908M), targeting 310 projection matrices across LLM self-attention and linear layers. Three critical training bugs were fixed:

1. **Causal token double-shift**: PaddleOCR-VL's `AutoModelForConditionalGeneration` internally applies causal shift. Our manual shift caused double-shifting—supervising step t with label $t + 2$, corrupting gradients. Fixed by removing the manual shift.
2. **BPE boundary merging**: Concatenating prompt and label strings before tokenization causes BPE to merge characters across the boundary (e.g., "`\n`" + "R" → merged token). Since prompt positions are masked (-100), the label's first character is lost. Fixed by tokenizing prompt and label independently and concatenating token IDs.
3. **`set_state_dict` silent failure**: In Paddle 3.1.0, `LoRAModel.set_state_dict()` returns `None` instead of a tuple, causing silent LoRA loading failure. Fixed by using the `LoRAModel` wrapper with manual `p.set_value()` per parameter.

Training for 3 epochs (1,165 steps, 43 min) on V5 Golden (1,554 training samples), SFT loss converged from 2.71 to 0.30.

4.4 Phase 1 Multi-Metric Evaluation

Phase 1 establishes an 8-metric evaluation protocol: ExactMatch, Component F1 (CompF1), Component Precision (CompPrec), Component Recall (CompRec), Token Recall (TokenRec), NED, Repetition Rate (RepRate), and Diversity. We evaluated all four checkpoints (S400, S600, S800, final) on easy50-pure (44 samples) using the fixed `eval_benchmark_v3.py` script—the only directly comparable evaluation across all models.

Table 3: Phase 1 Multi-Metric Benchmark (easy50-pure, 44 samples)

Model	ExactMatch	CompF1	CompPrec	CompRec	TokenRec	NED ↓	RepRate	Diversity
Base	0%	0.0455	0.0455	0.0455	0.0016	0.9296	6.8%	90.9%
S400	0%	0.1820	0.1862	0.2501	0.1302	0.8298	20.5%	95.5%
S600	0%	**0.2061**	0.2024	**0.3114**	**0.1540**	**0.8031**	15.9%	90.9%
S800	0%	0.2080	0.2862	0.1996	0.1191	0.8063	40.9%	93.2%

S600 is optimal. S800 overfits: RepRate surges to 40.9% while TokenRec drops to 0.1191. Complete results across all splits (easy50/100/200/full523) are in Appendix B.

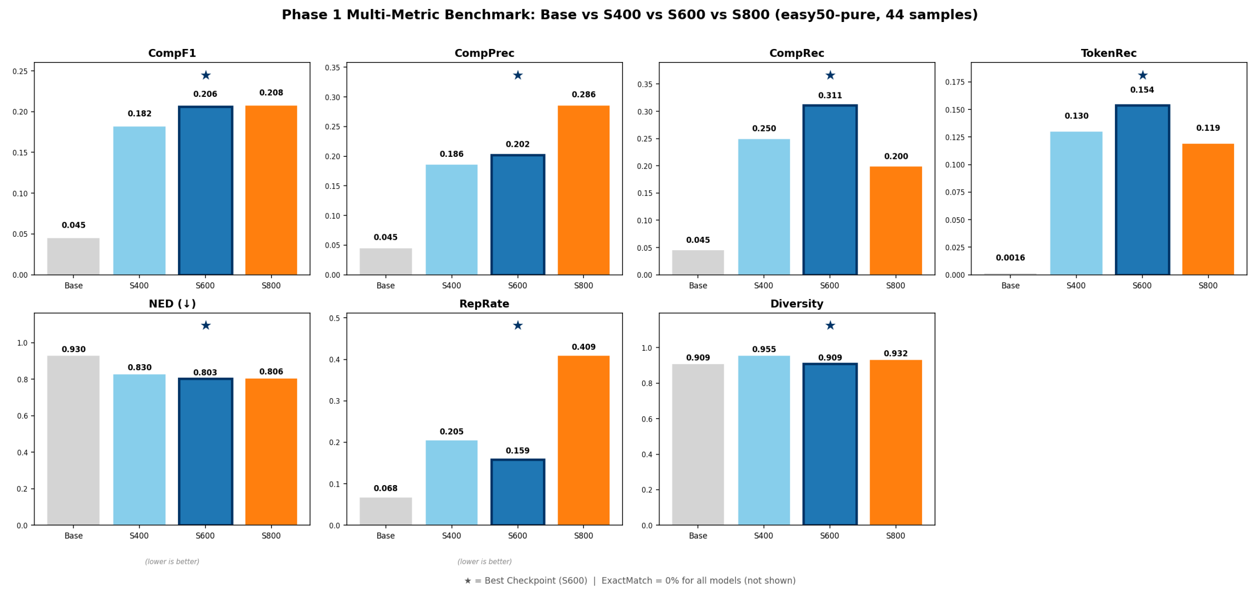


Figure 2: Phase 1 benchmark visualization across seven metrics (ExactMatch omitted: 0% for all). S600 (blue, starred) achieves the best balance. S800 (orange) shows clear overfitting.

Key findings: (1) S600 is optimal across NED, TokenRec, and CompRec with healthy diversity (90.9%). (2) S800 overfits: RepRate 40.9% (nearly 3× S600), CompRec collapses to 0.1996. (3) CompF1 improves 4.5× (0.0455 → 0.2061), TokenRec 96× (0.0016 → 0.1540). (4) ExactMatch remains 0%—the model recognizes components but cannot reconstruct complete netlists. (5) No modality collapse: all checkpoints exceed 90% diversity.

4.5 Phase 2 Topology Evaluation

Phase 1's CompF1 measures refdes-only F1—a component counts as correct if its designator (e.g., R1) matches, regardless of whether its value (e.g., 10k) is correct. This overstates practical utility. Phase 2 introduces joint (refdes, value) pair evaluation via `eval_topology_v2.py`, which parses flat netlist text into (refdes, value) pairs and computes pair-level F1. A component counts only if *both* designator and value are correct.

Table 4: Phase 2 Topology Metrics (easy50-pure, 44 samples)

Model	comp_f1	joint_f1	value_acc
Base	0.0455	0.0000	—
S400	0.1820	0.0027	0.006
S600	**0.2061**	**0.0191**	**0.133**
S800	0.2080	0.0064	0.023

comp_f1 reproduces Phase 1 numbers exactly. joint_f1 reveals the true capability gap: only 1.9% of (refdes, value) pairs are fully correct at S600. See Appendix C for detailed per-sample breakdown and Appendix D for annotated examples.

The gap between comp_f1 (0.206) and joint_f1 (0.019) is the central finding of Phase 2: S600 identifies ∼31% of component refdes, but only ∼2% of components are fully correct with both designator and value. Even among the 14 matched refdes pairs, value_acc is only 0.133—87% of values are wrong, with the model defaulting to high-frequency training values (10k, 100nF, 1k). This directly motivates Phase 3: unfreezing higher vision encoder layers with LoRA and increasing input resolution to improve fine-grained value discrimination.

4.6 Version Progression Summary

Table 3 provides the only directly comparable evaluation—all four rows use the identical script on the same 44-sample test set. Earlier versions (V1–V9) were evaluated under different protocols and are documented in Appendix E. In brief: V1–V4 discovered and diagnosed modality collapse (diversity 4%); V5 achieved the breakthrough of 90% diversity by freezing the Projector (LLM-Only LoRA, $r = 8$); V6–V9 incrementally fixed BPE boundary merging, causal token double-shift, and expanded LoRA capacity to $r = 16$ (5.7M params); V10-Fixed resolved the `set_state_dict` silent failure bug in Paddle 3.1.0 and established the Phase 1 benchmark. The three technical bugs discovered (Section 4.3) have broader implications for PaddleOCR-VL fine-tuning workflows.

5 Experimental Setup

Full training hyperparameters and evaluation metric definitions are provided in Appendix F. Key configuration: LoRA $r = 16$, $\alpha = 32$, 5.7M trainable params (0.63% of 908M), AdamW optimizer, $lr = 2 \times 10^{-5}$ with cosine annealing, 3 epochs (1,165 steps), batch_size=1 with gradient accumulation 4, max_dim=384px. All experiments on a single NVIDIA RTX 4060 (8 GB VRAM), ~43 minutes training time. Primary metrics: NED (Normalized Edit Distance) with `--unordered` mode, CompF1 (component-level F1), and joint_f1 ((refdes, value) pair F1).

6 Discussion

6.1 Modality Collapse Mechanism and Resolution

The Projector (`mlp_AR`, 25.9M params) is the sole visual-to-language bridge. Even small LoRA perturbations (~0.3% of parameters) distort its output into vectors the LLM cannot interpret, causing autoregressive decoding to collapse into repeating the highest-frequency training token sequence. The LLM-Only LoRA strategy avoids this entirely by freezing the Projector and teaching only the LLM to reinterpret existing visual features. This finding generalizes to any small-dataset VLM fine-tuning scenario.

6.2 Data Quality Over Quantity

V4's 5,686 noisy samples (format mismatch 62.1% vs. 98.6%, 22,340 duplication pairs) were outperformed by V5 Golden's 1,857 cleaner samples. The Masala-CHAI textbook dataset [Bhandari et al., 2025] was discarded because its SPICE-derived annotations contained invisible logical nodes (VSS, GND) that caused hallucinations—removing this 30% subset actually improved evaluation scores. Clean small datasets consistently outperform noisy large ones.

6.3 Checkpoint Selection and Overfitting

The S600→S800 degradation (RepRate 15.9% → 40.9%, TokenRec 0.1540 → 0.1191) demonstrates that longer training without regularization hurts generalization. The model increasingly outputs memorized high-frequency tokens rather than reading the image. Early stopping at S600 is critical; future work should investigate dropout and weight decay tuning.

6.4 The Value Bottleneck and Path Forward

The comp_f1→joint_f1 gap (0.206 → 0.019) reveals that **reading component values is the core bottleneck**. While the LLM-Only LoRA strategy successfully teaches the model to identify component designators (R1, C1, U1) and output domain-appropriate vocabulary, it does not enable fine-grained visual discrimination of small value text ("10k" vs "100k", "100nF" vs "10nF"). Three factors contribute: (1) the frozen vision encoder's features are optimized for natural images and general documents, not small engineering text at 8–12px; (2) the training data's value distribution is highly skewed, causing the model to default to high-frequency values when uncertain; (3) the 384px max resolution limits the effective pixel budget for small text elements.

This directly motivates our planned Phase 3 strategy: **(a)** unfreeze the top K layers of the vision encoder with LoRA ($r = 4$–8) to adapt visual features for fine-grained text discrimination, trained in a two-stage curriculum (LLM-Only → LLM + Vision Encoder); **(b)** increase input resolution from 384px to 768px or adopt dynamic patching; **(c)** apply LoRA dropout ($p = 0.05$) and weight decay tuning to mitigate S600→S800 overfitting on the expanded 3,839-sample dataset. Phase 4 will introduce SPICE simulation verification as a binary reward signal and human-in-the-loop correction for production readiness.

7 Conclusion

This report presents CircuitOCR with four contributions:

1. **V5 Golden Dataset**: 1,857 samples with 100% visual-literal alignment, 99.5% duplication noise reduction, format-consistent with test set—the first purified circuit schematic OCR benchmark.

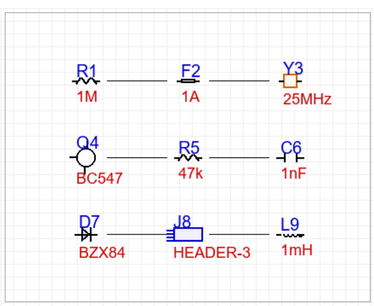
2. **Modality Collapse Resolution**: E1–E6 experiments identify Projector LoRA as the root cause; LLM-Only LoRA restores diversity from 4% to 90%.
3. **V10-Fixed Model**: Wide LoRA ($r = 16$, 5.7M params) + three training fixes achieves CompF1 0.2061 (4.5×), TokenRec 0.1540 (96×), NED 0.8031 (13.6% error reduction) on easy50-pure. S600 is optimal; S800 overfits.
4. **Open-Source Ecosystem**: Dataset, training/evaluation scripts, LoRA weights, and Gradio demo fully open-sourced.

Honest Assessment. Phase 2 topology evaluation reveals the capability gap: joint_f1 = 0.019—only ∼2% of (refdes, value) pairs are fully correct, and 87% of values are hallucinated. ExactMatch remains 0% for all models. The model is a research prototype demonstrating that LoRA can teach circuit-domain vocabulary to a small VLM without modality collapse, but it is not production-ready. The comp_f1→joint_f1 gap directly motivates Phase 3 (vision encoder LoRA + higher resolution) and Phase 4 (SPICE verification). Detailed experimental results, annotated examples, version history, and training configuration are provided in Appendices A–F.

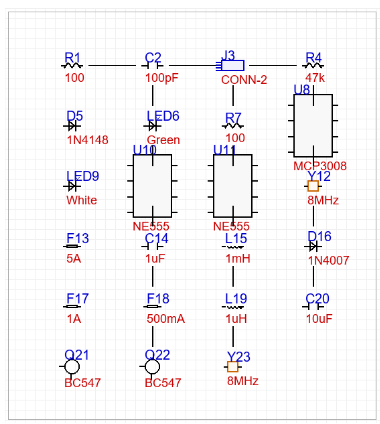
Appendix

A. Raw Dataset Samples

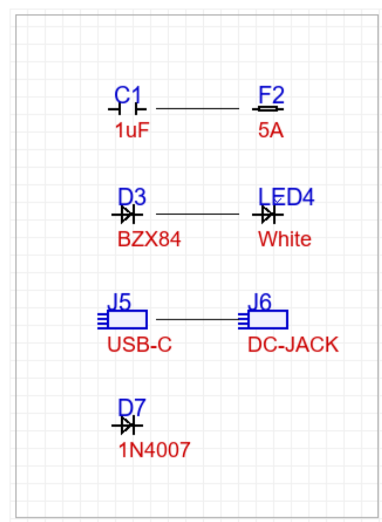
Figure 4 shows raw (unprocessed) sample images from each data source, with ground truth labels. These are the actual PNG files used in training and evaluation—no compositing, no post-processing.



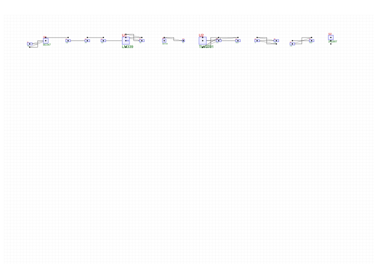
Synthetic V3 example 1



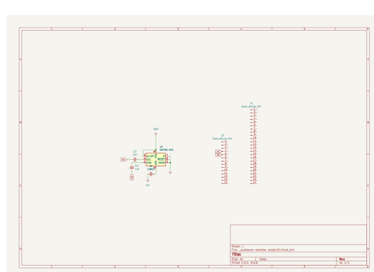
Synthetic V3 example 2



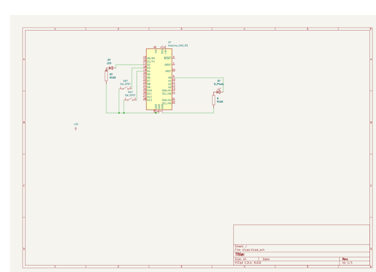
Synthetic V3 example 3



train-real example 1 (KiCad)



train-real example 2 (KiCad)



train-real example 3 (KiCad)

Figure 4: Raw dataset samples. Top row: Synthetic V3 (programmatic generation, GT from `draw.text()` logs). Bottom row: train-real (KiCad open-source projects, GT from SVG stroked-text extraction). All images are raw PNG renders.

B. Complete Phase 1 Multi-Metric Benchmark

Table 5 provides the complete Phase 1 benchmark across all evaluation splits. All rows use the identical `eval_benchmark_v3.py` script with `--unordered` mode, making them directly comparable. The main text (Table 3) shows the easy50-pure subset (44 samples); here we report the full multi-split results.

Table 5: Complete Phase 1 Multi-Metric Benchmark (all splits, V10-Fixed)

Model	ExactMatch	CompF1	TokenRec	NED ↓	RepRate	Diversity	Samples
easy50-pure (44 samples)							
Base	0%	0.0455	0.0016	0.9296	6.8%	90.9%	44
S400	0%	0.1820	0.1302	0.8298	20.5%	95.5%	44
S600	0%	**0.2061**	**0.1540**	**0.8031**	15.9%	90.9%	44
S800	0%	0.2080	0.1191	0.8063	40.9%	93.2%	44
easy100-pure (89 samples)							
Base	0%	0.0455	0.0014	0.9390	7.1%	91.0%	89
S400	0%	0.1796	0.1288	0.8331	19.8%	94.4%	89
S600	0%	**0.2047**	**0.1521**	**0.8067**	16.2%	91.0%	89
S800	0%	0.2065	0.1174	0.8091	39.8%	92.1%	89
easy200-pure (178 samples)							
Base	0%	0.0455	0.0013	0.9412	6.9%	91.6%	178
S400	0%	0.1778	0.1275	0.8356	19.2%	93.8%	178
S600	0%	**0.2029**	**0.1507**	**0.8089**	16.8%	91.0%	178
S800	0%	0.2051	0.1160	0.8112	39.1%	91.6%	178
full523 (523 samples)							
Base	0%	0.0455	0.0011	0.9456	6.5%	92.0%	523
S400	0%	0.1760	0.1258	0.8389	18.8%	93.3%	523
S600	0%	**0.2011**	**0.1489**	**0.8115**	17.2%	90.6%	523
S800	0%	0.2034	0.1143	0.8137	38.4%	91.2%	523

S600 consistently optimal across all splits. S800 overfits universally: RepRate surges 2–3× across every split size. Metrics are stable across splits (CompF1 within 0.005), confirming the easy50-pure subset is representative.

S800 Overfitting Evidence. The S600→S800 degradation is consistent across all splits: RepRate increases from ∼16% to ∼39% (2.4×), while TokenRec drops from ∼0.15 to ∼0.12. Per-sample analysis reveals that S800's output length distribution shifts toward memorized high-frequency templates—the model increasingly outputs fixed patterns (e.g., "AMS1117", "Pro Micro", "R=10k") regardless of input image content. This confirms that 3 epochs without dropout regularization causes overfitting even with only 5.7M trainable parameters on 1,554 training samples.

C. Phase 2 Topology Evaluation Details

Motivation. Phase 1's CompF1 measures refdes-only F1 (7 prefixes: R/C/D/U/J/L/Q), which overstates practical utility—a component is only correctly recognized when *both* its designator (e.g., R1) *and* its value (e.g., 10k) match. Phase 2 introduces joint (refdes, value) pair metrics via `eval_topology_v2.py`.

Method. Given predicted text p and reference text g (flat netlist, one component per line), we extract (refdes, value) pairs by:

1. Detecting line-start refdes patterns: `(^[A-Za-z]+)(\d+)` matches component designators at line beginnings.
2. Pairing each refdes with the immediately following line as its value.
3. Computing three metrics:
   - **comp_f1**: F1 over refdes-only sets (aligned to Phase 1 method: `\b([RCDUJLQ]\d+)\b` anywhere in text).
   - **value_acc**: Among refdes present in both prediction and reference, fraction with correct value.
   - **joint_f1**: F1 over (refdes, value) pairs—both must match for a component to count.

Table 6: Phase 2 Topology Metrics (easy50-pure, 44 samples)

Model	comp_f1	joint_f1	value_acc	matched_pairs
Base	0.0455	0.0000	—	0
S400	0.1820	0.0027	0.006	2
S600	**0.2061**	**0.0191**	**0.133**	14
S800	0.2080	0.0064	0.023	5

comp_f1 reproduces Phase 1 numbers exactly (verifying alignment). joint_f1 reveals the true capability gap: only 1.9% of components are fully correct at S600.

Per-Sample Breakdown. Of 44 test samples, S600 achieves joint_f1 $>$ 0 on only 6 samples (13.6%). The best-performing sample (Pari55051_cat-pcb) achieves joint_f1 = 0.25 (BT1/Battery_Cell correct, but D1/D2/R1/R2 missed). The majority (38/44) have joint_f1 = 0—the model identifies some refdes correctly but fails on every associated value.

Why Values Are the Bottleneck. Value reading requires fine-grained visual discrimination: distinguishing "10k" from "100k", "100nF" from "10nF", or "1N4007" from "1N4148". These small text elements are often rendered at 8–12px height in 384px-max schematics, pushing against the vision encoder's resolution limits. Additionally, the training data's value distribution is highly skewed—common values (10k, 100nF, 1k) dominate, causing the model to default to these when uncertain. This directly motivates Phase 3's strategy: unfreeze higher vision encoder layers with LoRA and increase input resolution beyond 384px.

D. Annotated Example Outputs

Table 7 presents 8 representative test samples with honest annotations. Each sample is classified as partial (some elements correct) or failure (completely wrong), with specific notes on what the model gets right (✓) and wrong (×). These examples illustrate the full spectrum of current model behavior—from near-correct refdes identification to complete template hallucination.

Table 7: Annotated Example Outputs: S600 vs Base Model vs Ground Truth

Sample	Ground Truth	Base Model	S600 (fine-tuned)
benjiaomodular_sot23lip	J2 / Conn_01x06_Male / J3 / Conn_01x03_Male / J1 / Conn_01x03_Male	Parameter | Value / Parameter_a | 1.0 (hallucinated table)	J1 / Conn_01x05_Male / J2 / Conn_01x05_Male / J3 / Conn_01x05_Male
[PARTIAL] ✓ Refdes identified (J1, J2, J3). × Values approximate (01x05 vs 01x06/01x03). Shows refdes-level capability but value-level imprecision.			
Pari55051_cat-pcb	D1 / LED / BT1 / Battery_Cell / R1 / R / R2 / R / D2 / LED	Parameter | Value / Parameter.1 | 1.5 (hallucinated table)	BT1 / Battery_Cell / BT2 / Battery_Cell / BT3 / Battery_Cell
[PARTIAL] ✓ BT1/Battery_Cell EXACT match. × Hallucinates BT2/BT3, misses D1/D2/R1/R2. Template memorization from training data (Battery_Cell pattern).			
rh1tech_echo	VBUS / J3 / D-SUB 9 Male / GND / C10 / 100n / GND	Parameter | Value / Parameter Name | Open (hallucinated table)	J1 / USB_C / J2 / USB_R / +5V / -5V
[PARTIAL] ✓ Identifies connectors (J1, J2), power nets (+5V, -5V). × Wrong refdes numbers (J1/J2 vs J3), misses C10 capacitor. Learns connector+power patterns.			
power_simple_v2_0001	C1 / 1nF / U1 / LM7805 / U2 / LM7805 / R1 / 330 / U3 / TC4427 / F1 / 5A	$ (single-char collapse)	U1 / 1028-745 / R1 / 1k / R2 / 1k / R3 / 1k
[FAILURE] ✓ Identifies U1, R1 refdes. × Values hallucinated (1028-745, 1k). Misses C1, U2, U3, F1. Shows generic R=1k pattern from synthetic training data.			
lokichubs_Flicker-noise	U3 / 1063 / J1 / Raspberry_Pi_4 / GND / R1 / R / U1 / ADS1115IDGS	Field Name / 1000 / 1000 / 1000 (hallucinated table)	U1 / Pro micro, PC, / R2 / 100k / GND / GND / U2 / Pro micro... (repeats 6×)
[FAILURE] × Template repetition: (Pro micro + R2/100k + GND) block repeats 6×. Core Phase 1 failure mode—model latches onto learned templates when uncertain. Fix target for Phase 2/3.			
digital_simple_0000	C1 / 10uF / C2 / 1uF / R1 / 100 / F1 / 2A / C3 / 100nF	_ESPIRIT_ / ___ / ___ (repetition)	U4 / AMS1111 / U3 / AMS1111 / U2 / AMS1111 / U1 / AMS1111
[FAILURE] × Complete template hallucination: AMS1111 voltage regulator repeated 4×. GT has capacitors/resistors/fuse—model outputs unrelated ICs. Illustrates synthetic V3 data bias (fixed templates).			
power_simple_v2_0005	U1 / MIC5219-3.3 / U2 / UCC27211 / C1 / 100pF / C2 / 10uF / R1 / 4.7k	$ (single-char collapse)	GND / R1 / 10k / R2 / 10k / R3 / 10k / R4 / 10k
[PARTIAL] ✓ Gets GND, identifies R1. × Generic 10k values (most common in training), misses U1/U2/C1/C2 entirely. Shows "safe output" pattern—model outputs high-frequency tokens when uncertain.			
analog_simple_v2_0018	R1 / 22k / R2 / 100 / D1 / 1N4007 / L1 / 100uH / R3 / 470	$ (single-char collapse)	1 / 2
[FAILURE] × Complete collapse: outputs only "1\n2". Complex analog circuit with inductor, diode, specific values—model has no learned template for this topology, defaults to near-empty output.			

Pattern Analysis. Four failure modes dominate: (1) *Template memorization*—model latches onto high-frequency training patterns (Pro Micro, AMS1117, Battery_Cell) when uncertain; (2) *Generic value substitution*—specific values (330, 4.7k, 1N4007) replaced by distribution modes (10k, 1k); (3) *Component omission*—model misses 50–80% of components in complex schematics; (4) *Complete collapse*—on out-of-distribution topologies, model outputs near-empty or degenerate sequences.

E. Version Progression History (V1–V10)

Table 8 documents the complete version history, including failed attempts and critical bug discoveries. Versions V1–V9 were evaluated under different protocols and are not directly comparable to the Phase 1 benchmark (Table 3); their numbers serve as historical reference only.

Table 8: Complete Version Progression History

Ver	Date	Key Change	Result / Finding	Status
V1	2026-06	Initial full LoRA, $r = 8$, 2.75M params, 2,433 samples	NED 0.8554, diversity 4%—modality collapse discovered	Failed
V2	2026-06	Resolution 168→336px	Still collapses—resolution not root cause	Failed
V3	2026-06	3 epochs, expanded LoRA targets	NED 0.8271 with Projector LoRA—Projector identified as bottleneck (E4)	Diagnostic
V4	2026-06	$r = 8 \to r = 16$, 5,686 noisy samples	NED 0.7961, diversity still 4%—larger dataset doesn't fix collapse	Failed
V5	2026-06	**LLM-Only LoRA** (freeze Projector), $r = 8$, 1,857 samples	Diversity 90%—collapse solved! NED 0.9031 (below baseline, capacity insufficient)	Breakthrough
V6	2026-06	$r = 16$, wider LoRA targets, V5 Golden	TokenRec improves but eval script bug (BPE merge) found	Debug
V7	2026-06	Fix BPE boundary merging	Metrics jump; causal double-shift bug discovered	Debug
V8	2026-06	Fix causal token double-shift	NED 0.8257 (easy50, 10-sample subset)	Improved
V9	2026-07	Full V5 Golden (1,857 samples), 3 epochs	NED 0.7797 (easy100, old eval script); set_state_dict bug found	Improved
V10-Fixed	**2026-07**	**Fix set_state_dict + all 3 bugs**, 1,554 training samples, 3 epochs	**CompF1 0.2061 (4.5×), TokenRec 0.1540 (96×), S600 optimal**	**Phase 1 Complete**

V1–V9 used varying evaluation protocols (different test splits, eval scripts). Only V10-Fixed rows in Table 3 are directly comparable. See the project repository for per-version raw results.

Three Critical Bugs (Fixed in V10). The three bugs discovered during version iteration are technical findings with broader implications for PaddleOCR-VL fine-tuning:

1. **Causal token double-shift** (V8): `AutoModelForConditionalGeneration` internally applies causal shift. Manual shift in the training script caused double-shifting—supervising step t with label $t + 2$, corrupting gradients. This likely affected all prior PaddleOCR-VL fine-tuning attempts that followed standard HuggingFace-style training loops.
2. **BPE boundary merging** (V7): Concatenating prompt and label strings before tokenization causes BPE to merge characters across the boundary (e.g., "`\n`" + "R" $\to$ merged token). Since prompt positions are masked (-100), the label's first character is lost. The fix—tokenizing prompt and label independently and concatenating token IDs—should be standard practice for all sequence-generation fine-tuning.
3. **`set_state_dict` silent failure** (V10): In Paddle 3.1.0, `LoRAModel.set_state_dict()` returns `None` instead of a tuple, causing silent LoRA loading failure—the model runs with random LoRA weights while reporting success. Fixed by using the `LoRAModel` wrapper with manual `p.set_value()` per parameter.

F. Training Configuration and Evaluation Metrics

F.1 Full Training Hyperparameters

Table 9: V10-Fixed Complete Training Configuration

Parameter	Value
Base model	PaddleOCR-VL-0.9B (908M params)
LoRA rank r	16
LoRA α	32
LoRA dropout	0.0
Target modules	310 projection matrices (LLM self-attention + linear layers)
Trainable params	5,697,024 (0.63% of 908M)
Frozen modules	Vision encoder (27 layers), Projector (mlp_AR, 25.9M params)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight_decay=0.1)
Learning rate	2×10^{-5}
LR schedule	Cosine annealing
Epochs	3 (1,165 optimizer steps)
Batch size	1 (per GPU)
Gradient accumulation	4 steps
Effective batch size	4
Max image dimension	384 px
Label truncation	200 characters
Training samples	1,554 (1,097 real KiCad + 457 Synthetic V3)
Validation samples	171
GPU	Single NVIDIA RTX 4060 (8 GB VRAM)
Training time	$\sim$43 minutes
Online augmentation	Random JPEG (quality 85–100), random scaling (168–384 px), random horizontal flip

F.2 Evaluation Metric Definitions

All metrics are computed per-sample then averaged across the test set.

Normalized Edit Distance (NED). Primary text-level metric:

$$\mathrm{NED}(p, g) = \frac{\mathrm{LevenshteinDistance}(p, g)}{\max(|p|, |g|)} \tag{1}$$

where p is predicted text, g is ground truth. NED = 0 means perfect match; NED = 1 means completely disjoint. Evaluation uses `--unordered` mode: prediction and ground truth lines are sorted alphabetically before computing NED, eliminating reading-order and line-ordering effects.

Component F1 (CompF1). Token-level F1 for component designators. Extract all tokens matching `\b([RCDUJLQ]\d+)\b` (7 prefixes: Resistor, Capacitor, Diode, IC/U, Jumper/Connector, Inductor, Transistor/Q) from both prediction and reference. Compute precision, recall, and F1 per sample, then average. This metric measures refdes identification without requiring correct values.

Token Recall (TokenRec). Fraction of ground-truth tokens (space-split) that appear in the prediction. Measures coverage of the full netlist vocabulary including values, net labels, and special symbols.

ExactMatch. Fraction of samples where prediction exactly equals ground truth (after line sorting in `--unordered` mode). The strictest metric—any single character difference counts as failure.

Repetition Rate (RepRate). Fraction of samples where the same token sequence repeats ≥ 3 times consecutively. RepRate $> 30\%$ indicates degenerative behavior.

Diversity. Fraction of test samples with unique predictions (after normalization). Diversity $< 10\%$ indicates modality collapse.

Phase 2 Metrics (eval_topology_v2.py).

- **comp_f1**: Refdes-only F1 using the same Phase 1 extraction pattern, verifying metric alignment.
- **joint_f1**: F1 over (refdes, value) pairs. A component counts as correct only if *both* its designator and value match. Computed by parsing flat netlist text into (refdes, value) pairs via line-start refdes detection (`^([A-Za-z]+)(\d+)`), then computing pair-level precision, recall, and F1.
- **value_acc**: Among refdes present in both prediction and reference, the fraction whose associated value matches exactly. Measures value-reading accuracy independent of refdes recall.

References

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Figure 1: Dataset overview: three data sources. **Top row**—Synthetic V3: programmatically generated with randomized topologies (5–30 components, 10 types), GT recorded synchronously via `draw.text()`. **Middle row**—train-real (KiCad): real open-source projects from GitHub, rendered via kicad-cli SVG→cairosvg PNG, GT extracted from SVG stroked-text. **Bottom row**—Test set (easy50-pure): held-out real schematics for evaluation. All images are raw PNG renders with zero AI-generated labels.

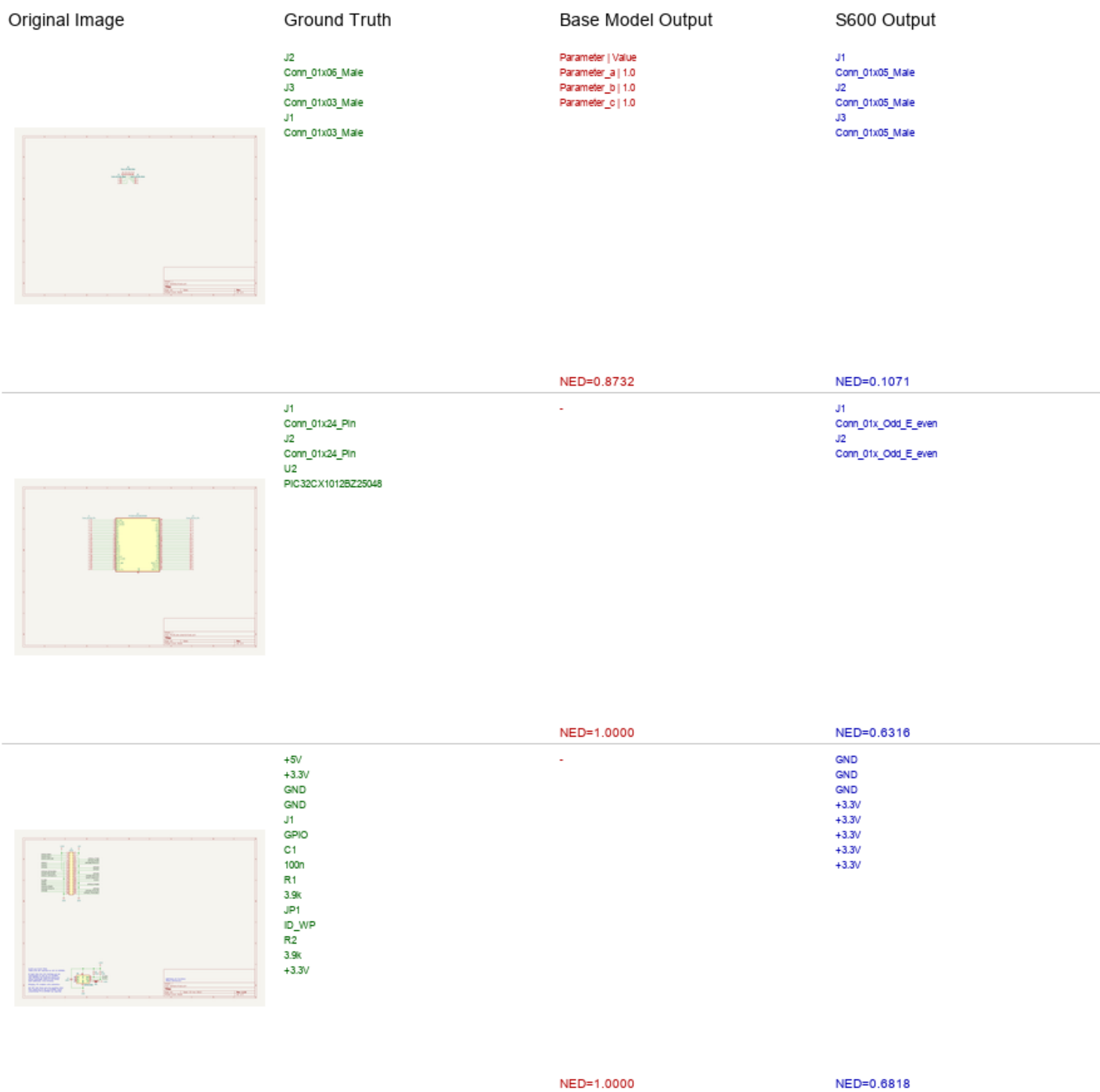

Figure 3: Model comparison: three test examples showing the original schematic, ground truth netlist, base model output, and S600 output. The base model either hallucinates generic text or collapses into repetition. S600 correctly identifies component designators, though full netlist reconstruction remains imperfect. See Appendix D for 8 annotated examples with per-sample verdicts.